# Vijaya Keerthi G


***DATA ENGINEER | 5+ YEARS***

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## PROFESSIONAL SUMMARY


**Technical Data Engineer** with **5+ years of experience** leading the architectural design, optimization, and end-to-end delivery of petabyte-scale distributed data platforms and modern AI infrastructure. Recognized for owning high-impact cloud migrations and streaming frameworks across Fortune 500 financial, healthcare, and automotive environments. Proven track record of managing cross-functional initiatives, implementing strict data contracts, and spearheading compute optimization strategies that slashed cloud infrastructure spend by up to 45%. Expert in building resilient Medallion architectures, production-grade GenAI/RAG ingestion pipelines, and low-latency event-driven systems utilizing advanced Python, PySpark, Apache Spark, Kafka, Snowflake, and AWS cloud-native tooling.


## TECHNICAL SKILLS

- **Programming Languages:** Python (Advanced/OOP), SQL (Expert - Window Functions, CTEs, Execution Plans), Scala, Java, Rust, Go, Bash
- **ETL, ELT & Data Integration Frameworks:** Custom ETL/ELT Pipeline Development, Source-to-Target Mapping (STTM), Incremental vs. Full Loads, Data Migration, Slowly Changing Dimensions (SCD Types 1/2/3), Data Cleansing & Deduplication, Fivetran, Airbyte, Stitch, Talend, Informatica PowerCenter
- **Big Data Processing & Distributed Compute:** Apache Spark (PySpark, SparkSQL), Apache Flink, Ray, dbt (Core, Cloud, Copilot), Pandas, Dask, MapReduce, Apache Beam, Trino, Starburst
- **GenAI, LLM Infrastructure & Knowledge Graphs:** Vector Databases (Pinecone, Milvus, Qdrant, Chroma, Weaviate), RAG (Retrieval-Augmented Generation) Ingestion Pipelines, LangChain, LlamaIndex, Embedding Architecture, Hugging Face API, Neo4j (Graph Databases/Knowledge Graphs)
- **MLOps & Feature Engineering:** Feast (Feature Store), Tecton, MLflow, DVC (Data Version Control), Weights & Biases, Databricks ML Runtime, AWS SageMaker Pipelines
- **Data Lakehouse & Enterprise Warehousing:** Snowflake, Databricks (Lakeflow, Delta Live Tables), Apache Iceberg, Delta Lake, Apache Hudi, AWS Redshift, Google Cloud BigQuery, Microsoft Fabric, HDFS
- **Streaming, Messaging & Eventing:** Apache Kafka, Confluent Cloud, AWS Kinesis, Spark Structured Streaming, Debezium (CDC), Segment, Apache Pulsar
- **Semantic & Activation Layers (Reverse ETL):** Cube.js, dbt Semantic Layer, Hightouch, Census
- **Orchestration & Workflow Automation:** Apache Airflow (Dynamic DAG Design), Dagster, Prefect, Argo Workflows, AWS Step Functions
- **DevOps, GitOps & Infrastructure:** Terraform (Infrastructure as Code), Pulumi, Docker, Kubernetes (K8s), Helm, Git, GitHub Actions, Jenkins, AWS, Google Cloud Platform (GCP), Microsoft Azure
- **Data Quality, Observability & Security:** Great Expectations, Monte Carlo, DataHub, Datafold, Databricks Unity Catalog, Apache Ranger, AWS IAM, Data Contracts, Schema Evolution
- **NoSQL, Graph & Relational Databases:** MongoDB, Cassandra, Redis, PostgreSQL, DynamoDB, Neo4j
- **Core Methodologies & Paradigms:** Dimensional Modeling (Star/Snowflake Schema), Medallion Architecture (Bronze/Silver/Gold), Data Mesh Architecture, Change Data Capture (CDC), Performance Optimization & Query Tuning, Memory Management (OOM Debugging), Partition Pruning, High Availability (HA) & Disaster Recovery (DR), PII Masking & Anonymization (GDPR/CCPA Compliance)

## PROFESSIONAL EXPERIENCE

**Data Engineer | Toyota** — ***April 2025 – Present | Dallas, TX***

- Led the architectural modernization of distributed ingestion frameworks using PySpark and AWS Glue, orchestrating the secure intake of 15TB+ daily manufacturing datasets into an Amazon S3 Medallion-structured data lakehouse.
- Owned the end-to-end delivery of low-latency streaming infrastructure utilizing Apache Kafka and Spark Structured Streaming, which accelerated vehicle telemetry processing speeds and slashed downstream analytical reporting latency by 45%.
- Directed cross-functional development initiatives to build a production-grade RAG pipeline using LangChain, converting messy enterprise documentation into high-performance Pinecone vector databases to automate contextual search engine capabilities.
- Spearheaded compute cluster optimization strategies by analyzing query execution plans and implementing custom partition pruning, eliminating chronic out-of-memory errors while reducing annualized AWS cloud infrastructure spend by 28%.
- Commanded the structural redesign of enterprise transactional databases into high-throughput Snowflake dimensional models, utilizing strict schema evolution data contracts to accelerate business intelligence time-to-insight metrics by four days.

**Environment:** Python, SQL, PySpark, Apache Spark, Spark Structured Streaming, Apache Airflow, Apache Kafka, AWS (S3, Glue, Athena, Redshift, Lambda), Snowflake, FastAPI, OpenAI API, LangChain, RAG, Pinecone, Docker, GitHub Actions, Jenkins, Parquet.

**Data Engineer | CVS** — ***May 2024 – April 2025 | Dallas, TX***

- Chaired the cloud migration team responsible for transitioning legacy database store procedures into cloud-native AWS architectures, successfully increasing high-volume healthcare ingestion velocities by 40% across internal operations.
- Championed enterprise data governance protocols by designing automated PII masking and tokenization microservices within AWS Lambda, ensuring absolute HIPAA compliance for millions of sensitive clinical, member, and provider profiles.
- Pioneered the development of highly scalable PySpark data parsing frameworks, which cleanly ingested multi-structured healthcare file formats like HL7 and FHIR into high-performance, optimized columnar Parquet storage structures.
- Managed database tuning initiatives across complex Amazon RDS PostgreSQL environments, restructuring clustering keys, indexing strategies, and analytical window functions to decrease operational dashboard query execution times by 35%.
- Instituted automated data quality frameworks using Great Expectations, establishing proactive runtime validation rules that trapped structural schema anomalies and successfully prevented corrupted financial metrics from impacting public reporting layers.

**Environment:** Python, SQL, PySpark, Apache Spark, Apache Airflow, Apache Kafka, AWS (S3, Glue, Athena, Redshift, Lambda, RDS PostgreSQL), Snowflake, FastAPI, LangChain, Vector Embeddings, HL7, FHIR, HIPAA, Parquet, JSON, Docker, Git, Jenkins, CI/CD.

**Data Engineer | JPMorgan Chase & Co.** — ***Jan 2020 – July 2022 | India,Hyderabad***

- Governed the engineering lifecycles of mission-critical financial processing systems built on Apache Spark, safely managing petabyte-scale investment banking data transactions with zero production workflow disruptions during high-volume quarters.
- Steered cross-team integration loops to deploy real-time reference data streams via Apache Kafka, eliminating core transaction settlement processing backlogs between front, middle, and back-office financial ledgers.
- Designed and implemented foundational warehouse blueprints using Star Schema concepts inside Amazon Redshift, drastically accelerating regulatory reporting compliance workflows for capital markets trading desks through streamlined aggregation pathways.
- Supervised the automation of ingestion pipelines from third-party global banking networks by writing robust, containerized Python workers scheduled programmatically through dynamically decoupled Apache Airflow workflow DAGs.
- Accountable for the implementation of end-to-end dual-ledger data reconciliation pipelines, verifying billions of historical records to guarantee complete mathematical data accuracy prior to final downstream financial executive sign-offs.

**Environment:** Python, SQL, PySpark, Apache Spark, Apache Airflow, Apache Kafka, AWS (S3, Glue, Athena, Redshift, Lambda, RDS PostgreSQL), Snowflake, FastAPI, Investment Banking, Trade Lifecycle, Star Schema, Dimensional Modeling, SCD, Docker, Git, Jenkins, CI/CD.

## ACADEMIC PROJECT

**AI-Powered Customer Insights Pipeline**

- Designed end-to-end ETL pipelines using Apache Airflow and PySpark to ingest, transform, and process structured and unstructured customer data, handling 2M+ records daily with optimized pipeline performance
- Built scalable data architecture using AWS S3 and Snowflake, improving query performance by 40% and enabling efficient data storage, retrieval, and analytics for business intelligence and machine learning workloads
- Implemented Retrieval-Augmented Generation (RAG) pipeline using LangChain and OpenAI API with vector embeddings, enabling semantic search and automated insight generation, reducing manual analysis effort by 60%

## EDUCATION

**Master of Science in Computer Science** — **Aug 2022 – May 2024**

University of North Texas, Denton, TX, USA